Results outline

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1 Wildchrokie

1.1 Growing season length and conditions

1. The shortest season by days was 2018 with 90 days and 2018, 2018 was the longest with 173 days.
2. The year with longest growing season length on average was 2019 with 141.5 days, and the shortest was 2020 with 135.9 days.
3. The hottest spring period of the three years (MAM) was 2020 with a mean mean temperature of 9.1°C, and the coldest, 2019 with a mean mean temperature of 8.7°C.
4. The hottest summer period of the three years (JJA) was 2020 with a mean mean temperature of 23.1°C, and the coldest, 2019 with a mean mean temperature of 22.4°C.
5. The driest spring period of the three years (MAM) was 2020 with a PDSI value of 1.3, and the wettest, 2019 with a PDSI value of 2.7.
6. The driest summer period (PDSI) of the three years (JJA) was 2020 with a PDSI value of -2.4, and the wettest, 2019 with a PDSI value of 4.
7. The spring period with the highest precipitation of the three years (MAM) was 2018 with a total precipitation of 362.6mm, and the lowest, 2020 with a total precipitation of 275.3mm.
8. The summer period with the highest precipitation of the three years (JJA) was 2019 with a total precipitation of 346mm, and the lowest, 2020 with a total precipitation of 199.1mm.

Table 1: Climate variables for each year and seasons.

Year	Season	PDSI	T_{max} (°C)	T_{mean} (°C)	T_{min} (°C)	PPT (mm)
2018	DJF	2.48	5.77	0.73	-4.30	301
2018	JJA	1.92	28.40	22.85	17.30	308
2018	MAM	2.44	14.33	8.90	3.47	363
2018	SON	3.31	17.17	12.13	7.10	483
2019	DJF	3.79	4.67	-0.12	-4.90	412
2019	JJA	4.02	28.13	22.38	16.63	346
2019	MAM	2.74	14.03	8.72	3.40	328
2019	SON	3.45	17.77	12.07	6.37	316
2020	DJF	1.29	6.37	1.78	-2.80	304
2020	JJA	-2.35	28.90	23.07	17.23	199
2020	MAM	1.32	14.57	9.05	3.53	275
2020	SON	-2.87	19.23	13.30	7.37	278

1.2 Phenology observations

1. Leafout dates across all years and species varied from 112 to 157 and budset date varied from 225 to 297.

2. On average, the earliest leafout date was in 2019 for all species (A. incana: 120.9, B alleghaniensis: 123, B. papyrifera: 123.3 and B. populifolia: 121). The latest leafout date for all species was in 2020 (A. incana: 148.7, B alleghaniensis: 141.3, B. papyrifera: 136.6 and B. populifolia: 133.8). No species was consistently the earliest nor the latest to leafout.

3. On average, the earliest budset date was in 2019 for A. incana (263.3) and B alleghaniensis (258), but in 2018 for B. papyrifera (266.8) and B. populifolia (266.3). The latest budset date for all species was in 2020 (A. incana: 280.9, B alleghaniensis: 264, B. papyrifera: 279.4 and B. populifolia: 282). No species was consistently the earliest nor the latest to set buds.

4. Across all species and for our three years, we found that PDSI and the total precipitation over the winter and spring were the only two significant climate predictors. Temperature did not affect leafout (Figure S3). Regression coefficients are present in Table S1

5. We found similar relationships with budset with the spring, summer and autumn PDSI and precipitation affecting this phenophase with no effect of temperature over these periods (Figure S3). Regression coefficients are present in Table S1

1.3 Ring widths

1. The ring widths of our species vary from 0.4mm to 13.5mm.

2. Ring widths for A. incana varied from 0.6mm to 7.8mm, from 0.4mm to 6.9mm for B. alleghaniensis, from 0.7mm to 6.4mm for B. papyrifera and from 0.8mm to 13.5mm for B. populifolia.

1.4 Model results

Our results show similar trends in response to the thermal (GDD) (Figure 2) versus the calendar growing season (GSL) (Figure 3), with a stronger response to GSL. Specifically, A. incana shows a marked shift in its response strength with longer GSL compared to GDD.

1. Though the effects are weak, all species appear to slightly grow more with increased GDD (A. incana (β = 0.3, UI_{90}[-0.1, 0.6]), B. alleghaniensis (β = 0.2, UI_{90}[-0.2, 0.6]) and B. papyrifera (β = 0.3, UI_{90}[-0.3, 0.9])), with the weakest effect being on B. populifolia (β = 0.1, UI_{90}[-0.4, 0.5]) (Figure 1).

2. A. incana ($\beta = 0.4$, UI_{90}[0.1, 0.6]) is the only species for which its 90% intervals don't overlap with zero suggesting that it grows more with longer growing season (GSL). B. alleghaniensis (β = 0.2, UI_{90}[-0.1, 0.5]) and B. papyrifera (β = 0.3, UI_{90}[-0.1, 0.8]) show a weaker response, but seem to grow more in response to longer seasons, with an even weaker response for B. populifolia (β = 0, UI_{90}[-0.3, 0.4]) (Figure 1).

3. Ring width decreased with a later start of season (SOS) for A. incana (β = -0.7, UI_{90}[-0.9, -0.4]) and B. populifolia (β = -0.4, UI_{90}[-0.8, -0.1]). The 90% UI overlap with 0 for B. alleghaniensis (β = -0.2, UI_{90}[-0.5, 0.1]) and B. papyrifera (β = -0.2, UI_{90}[-0.7, 0.3]) showing a weaker, but negative response to a later SOS (Figure 1).

4. Ring width decreased with a later end of season (EOS) for A. incana (β = -0.3, UI_{90}[-0.5, 0]). B. populifolia showed a weaker response but appears to also grow less with a later EOS (β = -0.2, UI_{90}[-0.5, 0.1]). B. alleghaniensis (β = 0, UI_{90}[-0.2, 0.3]) and B. papyrifera (β = 0.1, UI_{90}[-0.3, 0.5]) didn't respond to EOS. (Figure 1).

5. For the provenance with the GDD model, the estimates didn't shift much across climate predictors. We detected a weak growth difference between G. College (α_{site} = -0.4, UI_{90}[-2.1, 1.2]), S. Hypolyte (α_{site} = -0.1, UI_{90}[-1.8, 1.6]), W. Mountains (α_{site} = 0.5, UI_{90}[-1.2, 2.2]) and H. Forest (α_{site} = = -0.8, UI_{90}[-2.5, 0.9]) (Figure 1)

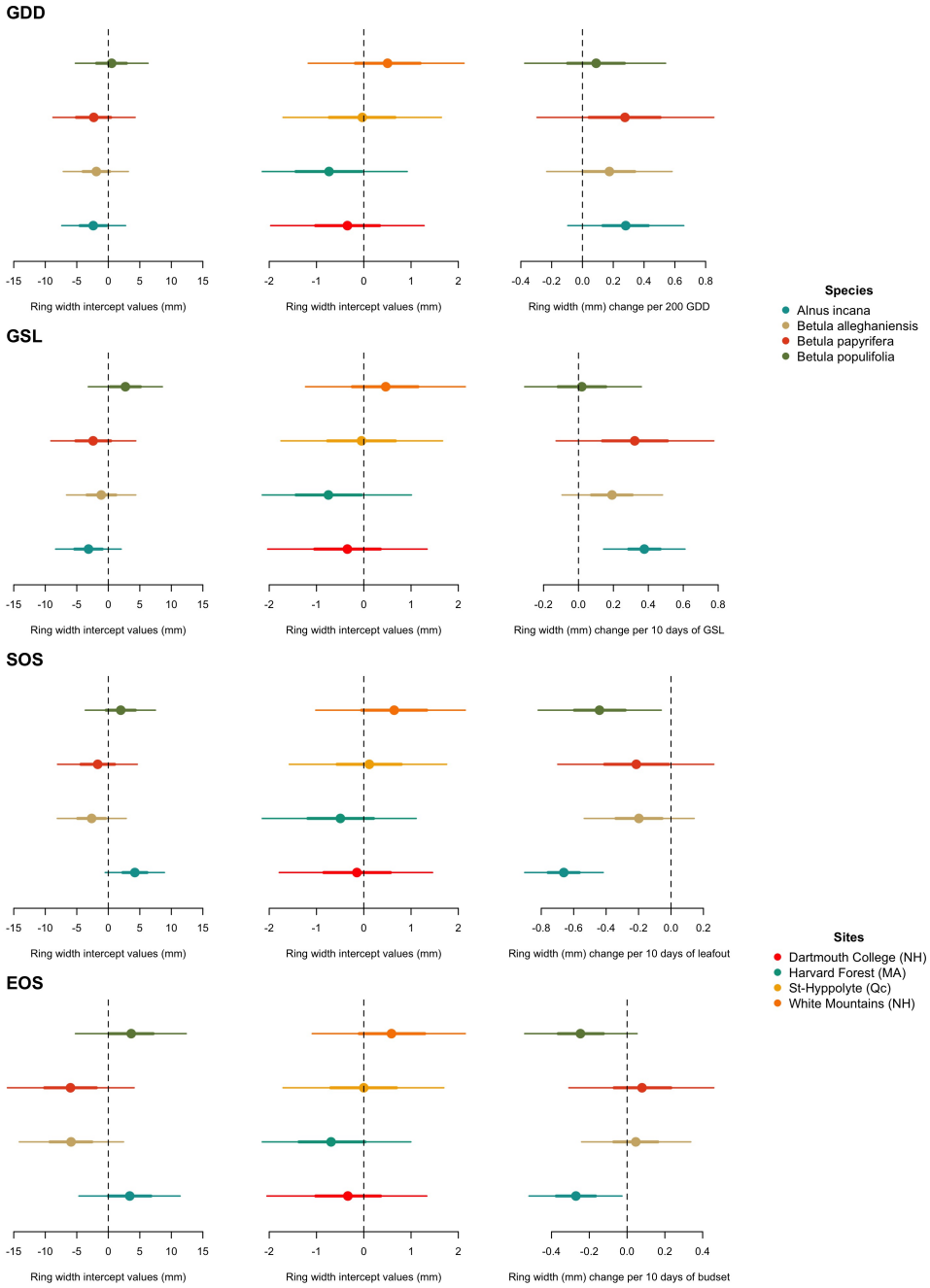

Figure 1: Model parameters across the different model predictors. GDD represent the accumulated growing degree days between leafout and budset, GSL the number of days between these phases. SOS represent the start of season, which is leafout and EOS, the end of season through budset. All predictors were fit using the same model and same observation, with only the independant variable changing (see Figure S1 for a comparison between two different data set length for SOS, and Figure S2, for end of EOS). The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Supplemental results

1. Fitting start of season (leafout) adds 12 rows of data out of the 227 when removing both NA entries for leafout and budset, and 3 rows of data when fitting the full dataset of end of season (budset).

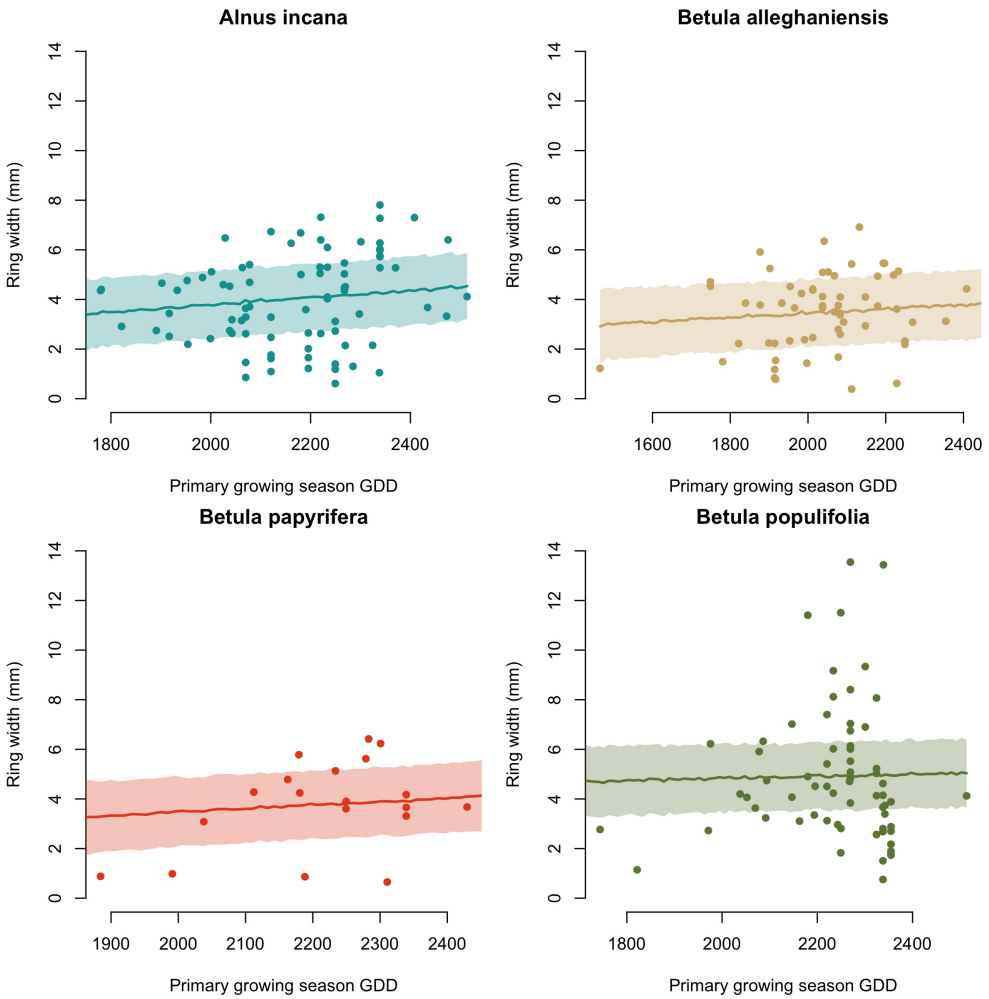

Figure 2: Ring width responses to total growing degree days (GDD) accumulated between the leafout and budset date. The length of the growing season is calculated between the leafout and budset date. The shaded areas represent the 50% confidence intervals of the posterior distribution with measurement error, and the lines are their means. The dots represent the empirical data

2. Fitting with the maximum number of rows of data didn't change the parameter estimates by much for SOS (Figure S1) and EOS (Figure S2).

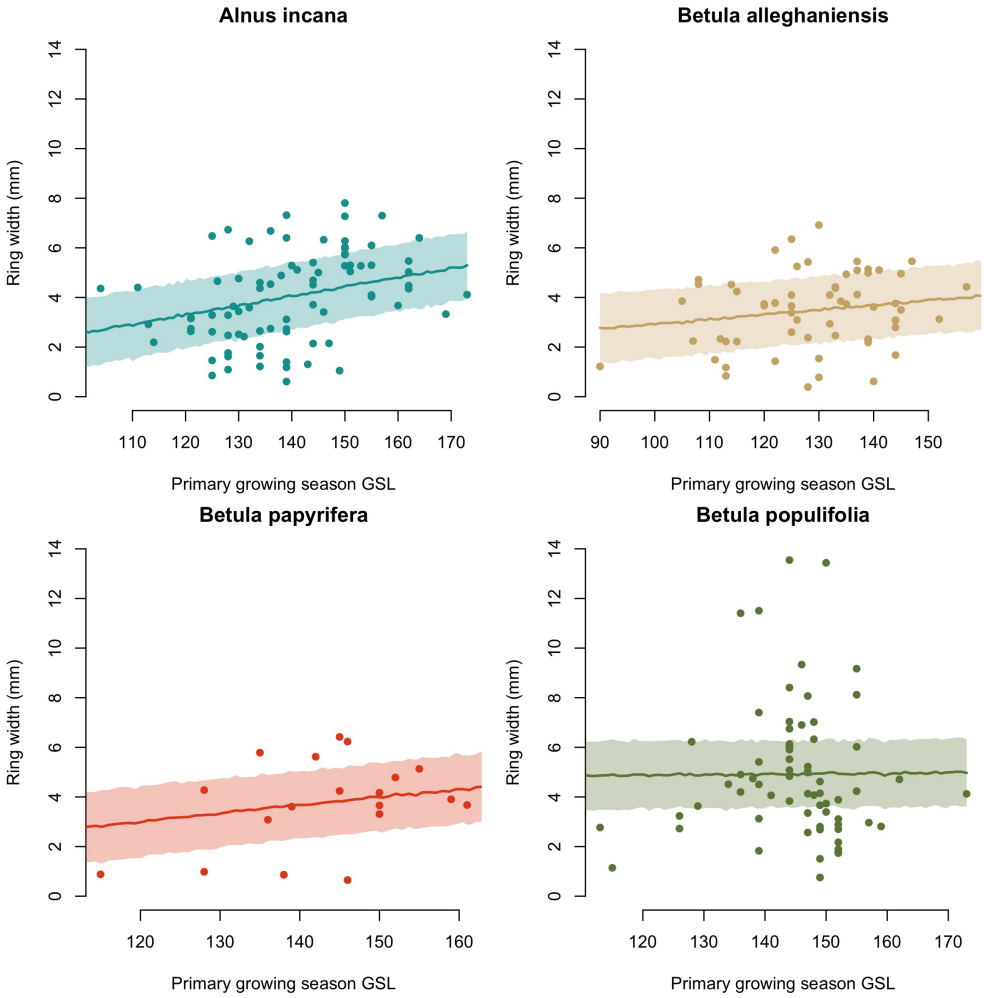

Figure 3: Ring width responses to the length of the primary growing season (days). The length of the growing season is calculated between the leafout and budset date. The shaded areas represent the 50% confidence intervals of the posterior distribution with measurement error, and the lines are their means. The dots represent the empirical data

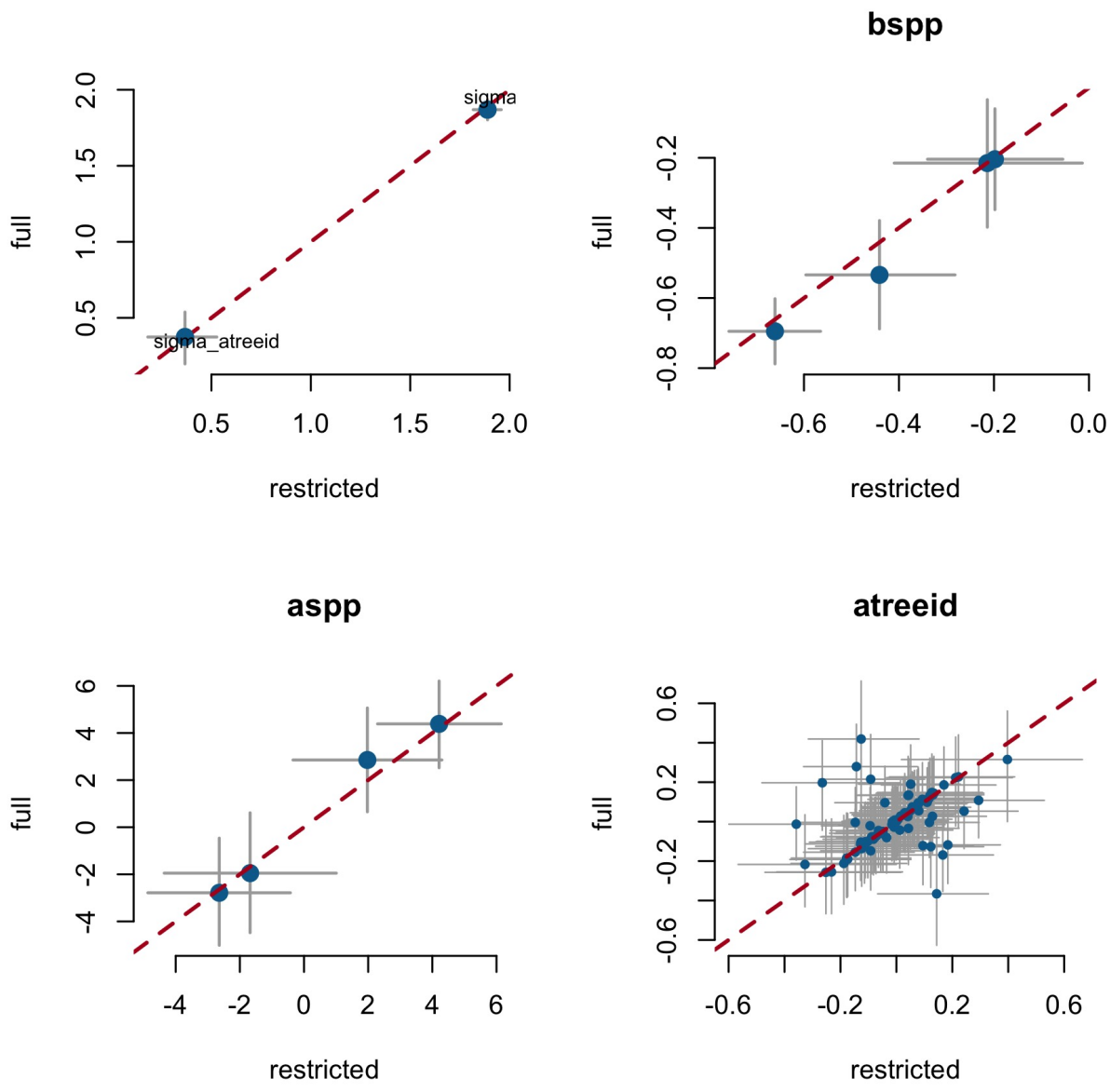

Figure S1: Full vs most restricted rows of data for the model with SOS as the predictor. Each pannel represent the different parameters used to fit the models.

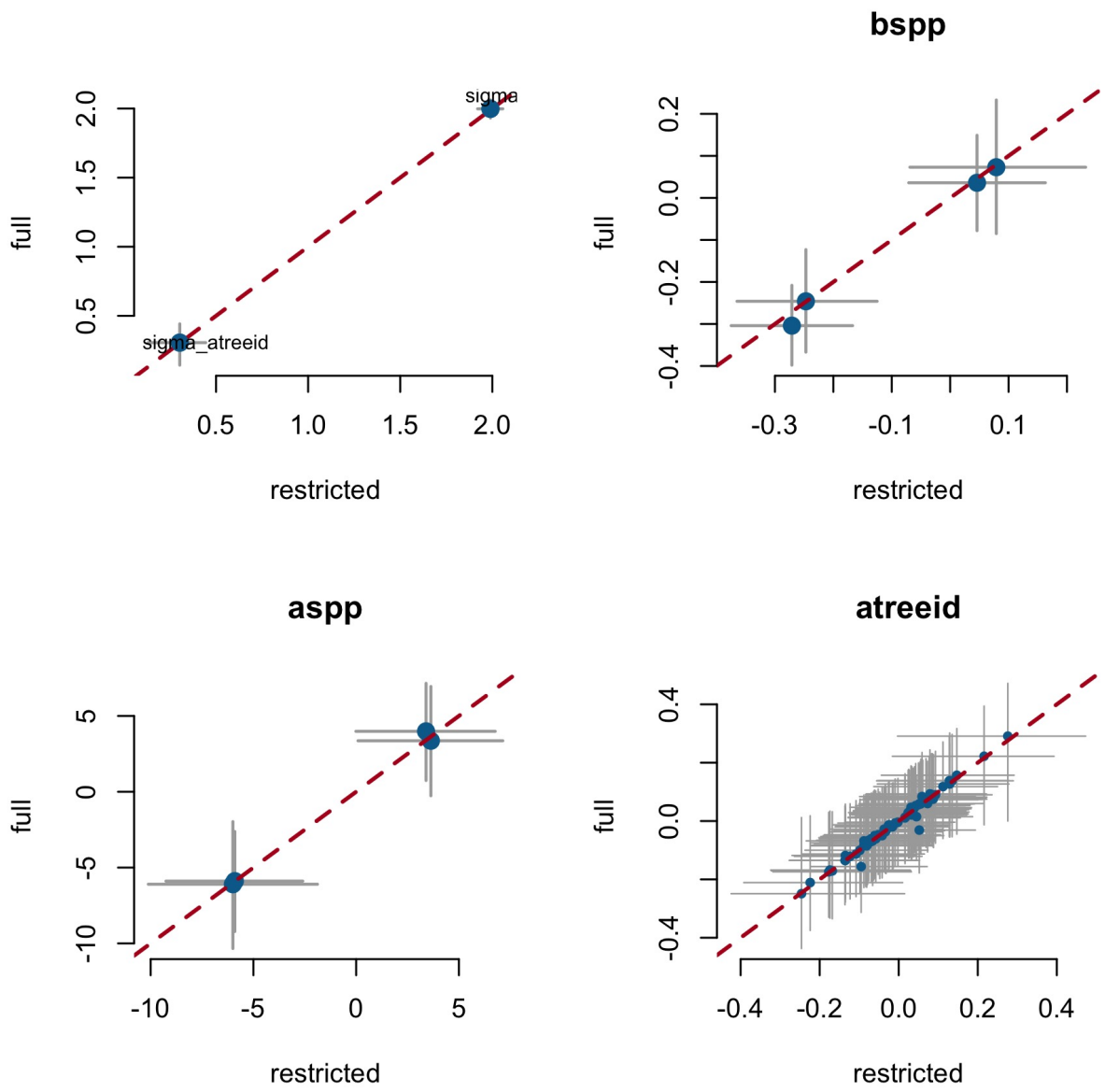

Figure S2: Full vs most restricted rows of data for the model with EOS as the predictor. Each pannel represent the different parameters used to fit the models.

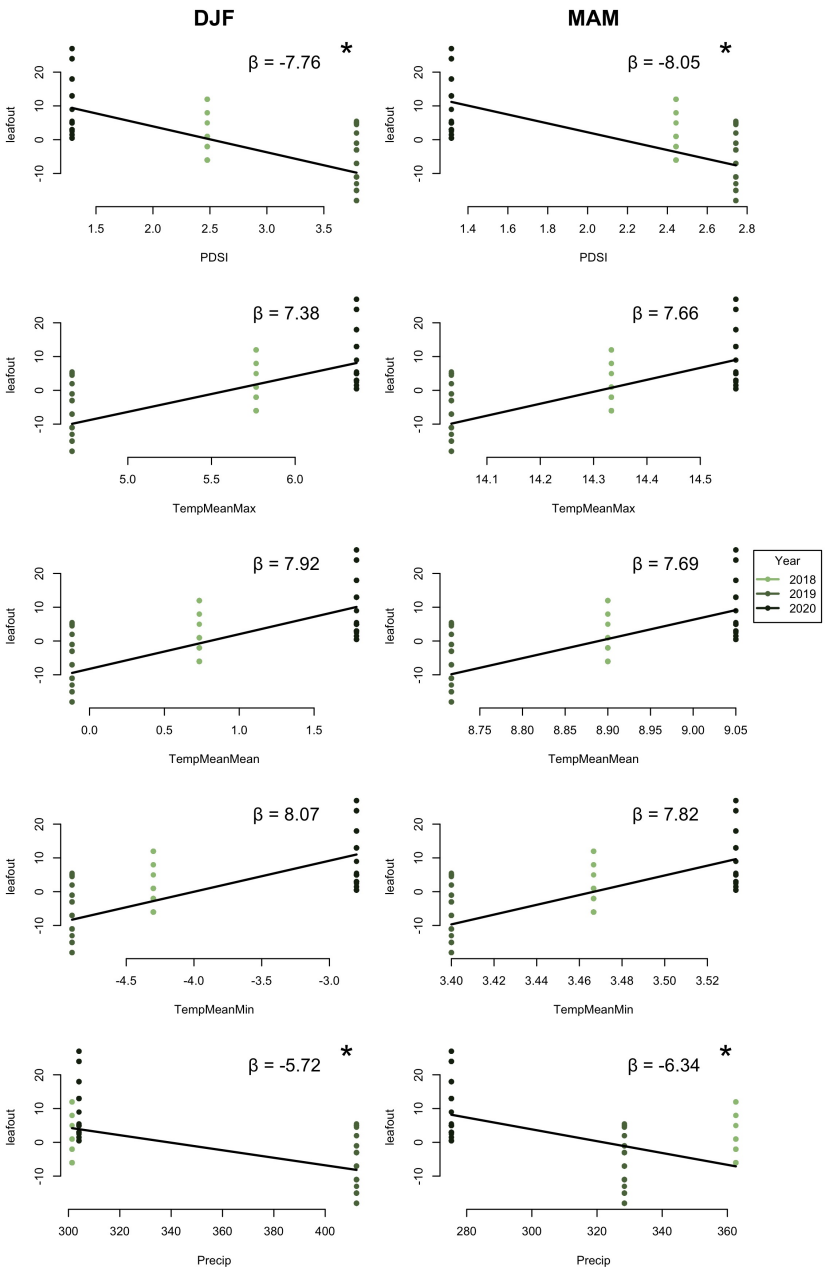

Figure S3: Leafout as a function of different climate predictors across winter (DJF) and spring (MAM). DJF is for December, January and February and MAM is for March, April and May. PDSI represents the Palmer Drought Severity Index averaged for each period. TempMeanMax, TempMeanMean and TempMeanMin are the averaged maximum, mean and mininimum monthly temperature over the 3-month periods. Precip is the total precipitation (mm) received over the 3-month periods.

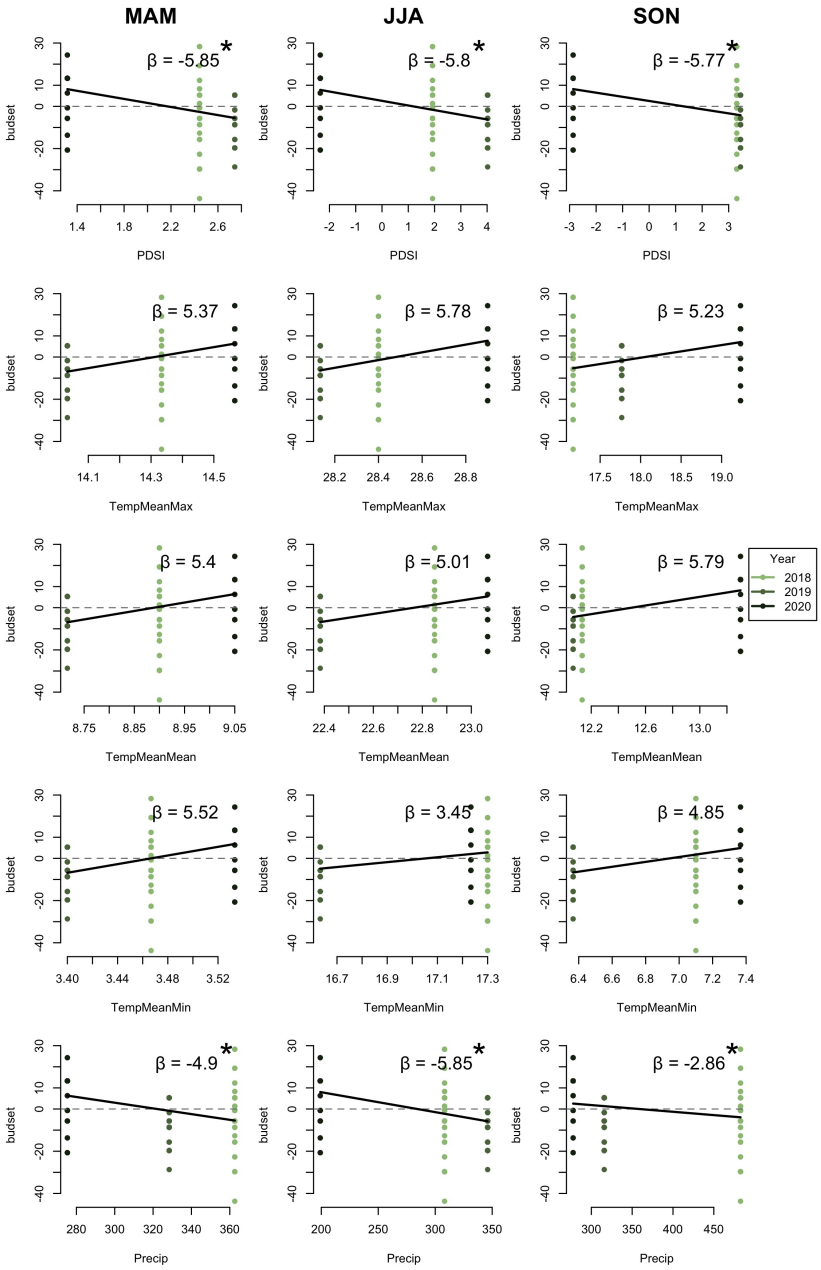

Figure S4: Budset as a function of different climate predictors across spring (MAM), summer (JJA) and autumn (SON). MAM is for March, April and May, JJA is for June, July and August and SON is for September, October and November. PDSI represents the Palmer Drought Severity Index averaged for each period. TempMeanMax, TempMeanMean and TempMeanMin are the averaged maximum, mean and mininimum monthly temperature over the 3-month periods. Precip is the total precipitation (mm) received over the 3-month periods.

Table S1: Regression coefficients for the lmer model with years as random effects.

Predictor	Season	Slope	Std. Error	t-value	p-value
leafout					
PDSI	DJF	-7.76	2.12	-3.66	0.17
PDSI	MAM	-8.05	0.69	-11.66	0.05
TempMeanMax	DJF	7.38	3.18	2.33	0.26
TempMeanMax	MAM	7.66	2.45	3.12	0.20
TempMeanMean	DJF	7.92	1.42	5.58	0.11
TempMeanMean	MAM	7.69	2.34	3.28	0.19
TempMeanMin	DJF	8.07	0.46	17.55	0.00
TempMeanMin	MAM	7.82	1.90	4.12	0.15
Precip	DJF	-5.72	5.69	-1.01	0.50
Precip	MAM	-6.34	5.18	-1.22	0.44
budset					
PDSI	MAM	-5.85	0.74	-7.92	0.00
PDSI	JJA	-5.80	0.77	-7.55	0.07
PDSI	SON	-5.77	1.05	-5.49	0.11
TempMeanMax	MAM	5.37	2.26	2.38	0.25
TempMeanMax	JJA	5.78	0.89	6.52	0.09
TempMeanMax	SON	5.23	2.76	1.90	0.31
TempMeanMean	MAM	5.40	2.18	2.48	0.24
TempMeanMean	JJA	5.01	2.96	1.69	0.34
TempMeanMean	SON	5.79	0.89	6.53	0.09
TempMeanMin	MAM	5.52	1.86	2.96	0.20
TempMeanMin	JJA	3.45	4.72	0.73	0.60
TempMeanMin	SON	4.85	3.23	1.50	0.37
Precip	MAM	-4.90	3.34	-1.47	0.38
Precip	JJA	-5.85	0.74	-7.91	0.00
Precip	SON	-2.86	5.23	-0.55	0.68